

Lecture 8

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Convention. The ground set is N , and $M = (N, \mathcal{I})$ is a matroid with rank function r and set of bases $\mathcal{B}(M)$. For $x \in \mathbb{R}^N$ and $S \subseteq N$, write $x(S) = \sum_{i \in S} x_i$. The incidence vector of S is χ^S , and $B - i + j = (B \setminus \{i\}) \cup \{j\}$. Unless stated otherwise, $f : 2^N \rightarrow \mathbb{R}$ is submodular; neither monotonicity nor nonnegativity is needed for the rounding arguments.

1 The Multilinear Relaxation

The independence and base polytopes of M are

$$\begin{aligned} P(M) &= \{x \in \mathbb{R}_{\geq 0}^N : x(S) \leq r(S) \text{ for every } S \subseteq N\}, \\ B(M) &= \{x \in P(M) : x(N) = r(N)\} = \text{conv}\{\chi^B : B \in \mathcal{B}(M)\}. \end{aligned}$$

The singleton rank inequalities imply $x_i \leq 1$, so $P(M) \subseteq [0, 1]^N$.

For $x \in [0, 1]^N$, let $R(x)$ contain every $i \in N$ independently with probability x_i . The *multilinear extension* of f is

$$F(x) = \mathbb{E}[f(R(x))] = \sum_{S \subseteq N} f(S) \prod_{i \in S} x_i \prod_{i \notin S} (1 - x_i). \quad (1)$$

Continuous algorithms first find a good fractional point in $P(M)$ and then need to round it without losing the value promised by F .

1.1 Curvature of the multilinear extension

For distinct i, j , differentiating (1) gives

$$\frac{\partial^2 F}{\partial x_i \partial x_j}(x) = \mathbb{E}[f(S + i + j) - f(S + i) - f(S + j) + f(S)] \leq 0, \quad (2)$$

where S is sampled from $N \setminus \{i, j\}$ using the coordinates of x . The inequality is exactly submodularity. Moreover, $\partial^2 F / \partial x_i^2 = 0$ because F is multilinear.

Lemma 1.1 (Ray concavity and swap convexity). *Let $x \in [0, 1]^N$.*

- (i) *If $d \in \mathbb{R}_{\geq 0}^N$, then $t \mapsto F(x + td)$ is concave on every interval on which $x + td \in [0, 1]^N$.*
- (ii) *If $i \neq j$, then $t \mapsto F(x + t(\mathbf{e}_i - \mathbf{e}_j))$ is convex on every such interval.*

Proof. For (i), the second derivative is $d^\top \nabla^2 F(x + td) d \leq 0$: the diagonal terms vanish, the off-diagonal Hessian entries are nonpositive by (2), and $d_i d_j \geq 0$. For (ii),

$$\frac{d^2}{dt^2} F(x + t(\mathbf{e}_i - \mathbf{e}_j)) = -2 \frac{\partial^2 F}{\partial x_i \partial x_j}(x + t(\mathbf{e}_i - \mathbf{e}_j)) \geq 0.$$

□

The second property is the analytic engine behind both rounding schemes in this lecture.

Theorem 1.2 (Dependent rounding goal [8]). *Given $x \in P(M)$, one can sample in polynomial time an independent set $R \in \mathcal{I}$, using no oracle access to f , such that*

$$\mathbb{E}[\chi^R] = x, \quad \mathbb{E}[f(R)] \geq F(x).$$

Thus the rounding incurs no loss relative to the multilinear relaxation.

Remark (Historical context). Pipage rounding originated in work of Ageev and Sviridenko on coverage and cut problems [1, 2]. Călinescu, Chekuri, Pál, and Vondrák applied it to submodular maximization over a matroid and developed the continuous-greedy framework [4, 5]; see also Vondrák's value-oracle formulation [10]. Swap rounding and its general exchange framework are due to Chekuri, Vondrák, and Zenklusen [7, 8].

2 A Mean-Preserving Swap Lemma

The following lemma isolates the common proof behind pipage and swap rounding.

Lemma 2.1 (Swap submartingale). *Let $(X_t)_{t=0}^T$ be a stochastic process in $[0, 1]^N$. Suppose that, conditionally on the history through time t , there are distinct coordinates i, j and a random scalar ξ_t such that*

$$X_{t+1} = X_t + \xi_t(\mathbf{e}_i - \mathbf{e}_j), \quad \mathbb{E}[X_{t+1} \mid X_t, i, j] = X_t.$$

Then

$$\mathbb{E}[F(X_{t+1}) \mid X_t] \geq F(X_t), \quad \text{and hence} \quad \mathbb{E}[F(X_T)] \geq F(X_0).$$

Also (X_t) is a vector-valued martingale, so $\mathbb{E}[X_T] = \mathbb{E}[X_0]$.

Proof. The conditional mean identity gives $\mathbb{E}[\xi_t \mid X_t, i, j] = 0$. By Lemma 1.1, $g(s) = F(X_t + s(\mathbf{e}_i - \mathbf{e}_j))$ is convex on the feasible interval. Jensen's inequality therefore gives

$$\mathbb{E}[F(X_{t+1}) \mid X_t, i, j] = \mathbb{E}[g(\xi_t) \mid X_t, i, j] \geq g(0) = F(X_t).$$

Take another conditional expectation to remove i, j , and then iterate. The coordinate expectation follows directly from the mean identity. $\square$

The lemma explains why the algorithms are oblivious to f : their random choices only need to preserve the current vector in expectation.

3 Randomized Pipage Rounding

Pipage rounding starts at $x \in B(M)$ and repeatedly moves to an endpoint of a feasible swap line. Each endpoint lies on a lower-dimensional face, so the process eventually reaches a vertex of the base polytope.

3.1 Tight sets and feasible swap directions

A set $S \subseteq N$ is *tight* at $x \in P(M)$ if $x(S) = r(S)$.

Lemma 3.1 (Uncrossing tight sets). *If A and B are tight at $x \in P(M)$, then $A \cap B$ and $A \cup B$ are tight at x .*

Proof. Submodularity of r , feasibility of x , and modularity of $x(\cdot)$ give

$$\begin{aligned} x(A) + x(B) &= r(A) + r(B) \\ &\geq r(A \cap B) + r(A \cup B) \\ &\geq x(A \cap B) + x(A \cup B) = x(A) + x(B). \end{aligned}$$

Every inequality is therefore an equality. □

For fractional coordinates i, j , define the largest feasible steps

$$\delta^+(x; i, j) = \min \left\{ x_j, \min_{\substack{S \subseteq N \\ i \in S, j \notin S}} (r(S) - x(S)) \right\}, \quad (3)$$

$$\delta^-(x; i, j) = \min \left\{ x_i, \min_{\substack{S \subseteq N \\ j \in S, i \notin S}} (r(S) - x(S)) \right\}. \quad (4)$$

Thus $x + \delta^+(\mathbf{e}_i - \mathbf{e}_j)$ and $x - \delta^-(\mathbf{e}_i - \mathbf{e}_j)$ are the two endpoints of the feasible interval. The terms x_j and x_i in (3)–(4) enforce nonnegativity; omitting them would make the step-size formula incomplete. The inner minima are instances of submodular function minimization with membership constraints.

If $x \in B(M)$ is nonintegral, then it has at least two fractional coordinates because $x(N) = r(N)$ is an integer. Choose an inclusion-minimal tight set T containing at least two fractional coordinates, and choose fractional $i, j \in T$. No tight set separates i from j . Indeed, if a tight set S contains i but not j , then $S \cap T$ is a proper tight subset of T . It contains fractional i , and it cannot contain exactly one fractional coordinate because its sum $x(S \cap T) = r(S \cap T)$ is integral. This contradicts the minimality of T . The reverse separation is symmetric. Consequently both step sizes in (3)–(4) are positive.

Algorithm 1 Randomized pipage rounding

Input: A point $x \in B(M)$.

while x is not integral **do**

Choose an inclusion-minimal tight set T containing at least two fractional coordinates.

Choose distinct fractional $i, j \in T$.

$\delta^+ \leftarrow \delta^+(x; i, j)$ and $\delta^- \leftarrow \delta^-(x; i, j)$.

$x^+ \leftarrow x + \delta^+(\mathbf{e}_i - \mathbf{e}_j)$ and $x^- \leftarrow x - \delta^-(\mathbf{e}_i - \mathbf{e}_j)$.

With probability $\delta^-/(\delta^+ + \delta^-)$ set $x \leftarrow x^+$;

otherwise set $x \leftarrow x^-$.

end while

return the unique base R with $x = \chi^R$.

Theorem 3.2 (Pipage rounding). *For $x_0 \in B(M)$, Algorithm 1 returns in polynomial time a random base $R \in \mathcal{B}(M)$ such that*

$$\mathbb{E}[\chi^R] = x_0, \quad \mathbb{E}[f(R)] \geq F(x_0).$$

Proof. Both endpoints in every iteration lie in $B(M)$. Moreover,

$$\begin{aligned} \mathbb{E}[x_{\text{new}} \mid x] &= \frac{\delta^-}{\delta^+ + \delta^-} (x + \delta^+ (\mathbf{e}_i - \mathbf{e}_j)) \\ &\quad + \frac{\delta^+}{\delta^+ + \delta^-} (x - \delta^- (\mathbf{e}_i - \mathbf{e}_j)) = x. \end{aligned}$$

Thus the iterates satisfy Lemma 2.1.

It remains to prove progress. Every constraint tight at x contains either both i, j or neither, so the swap line lies in the affine hull of the minimal face of $B(M)$ containing x . Moving to either maximal endpoint makes a coordinate constraint or a previously slack rank constraint tight. Hence the endpoint belongs to a proper subface. The minimal-face dimension drops in every iteration, so at most $\dim B(M) \leq |N| - 1$ iterations are needed. At termination x is an integral point of $B(M)$, hence $x = \chi^R$ for a base R .

The martingale part of Lemma 2.1 gives the stated marginals, and its submartingale part gives $\mathbb{E}[f(R)] = \mathbb{E}[F(\chi^R)] \geq F(x_0)$. Tight sets and step lengths can be found in polynomial time by submodular minimization, so the full procedure is polynomial. $\square$

3.2 From the base polytope to the independence polytope

The direct algorithm requires a point of $B(M)$; the statement for $P(M)$ follows from a standard lift. Let $r = r(N)$, introduce a set D of r dummy elements, and let

$$\widetilde{M} = \text{Trunc}_r(M \oplus U_{r,D}).$$

For any $x \in P(M)$, distribute mass $r - x(N)$ among the dummy coordinates, with every dummy coordinate in $[0, 1]$, to obtain $\tilde{x} \in B(\widetilde{M})$. Indeed, every lifted rank inequality follows from $x(A) \leq r_M(A)$, $\tilde{x}(Q) \leq |Q|$ for $Q \subseteq D$, and $\tilde{x}(N \cup D) = r$. Extend the objective by $\tilde{f}(S) = f(S \cap N)$, which remains submodular. Pipage-round $\tilde{x}$ and discard the dummies. The resulting set is independent in M , has original-element marginals x , and has expected value at least $F(x)$. This proves Theorem 1.2 using pipage rounding.

4 Randomized Swap Rounding

Swap rounding starts not from rank inequalities but from a convex decomposition

$$x = \sum_{\ell=1}^m \lambda_\ell \chi^{B_\ell}, \quad B_\ell \in \mathcal{B}(M), \quad \lambda_\ell > 0, \quad \sum_{\ell=1}^m \lambda_\ell = 1. \quad (5)$$

By Carathéodory's theorem one may take $m \leq |N|$. Cunningham's algorithm finds such a decomposition using $O(|N|^6)$ independence-oracle calls [6]. Simply sampling B_ℓ with probability λ_ℓ preserves the marginals, but it need not guarantee $\mathbb{E}[f(R)] \geq F(x)$. The bases must instead be merged through mean-preserving exchanges.

Lemma 4.1 (Symmetric basis exchange [3]). *If $B_1, B_2 \in \mathcal{B}(M)$ and $i \in B_1 \setminus B_2$, then there is $j \in B_2 \setminus B_1$ such that both $B_1 - i + j$ and $B_2 - j + i$ are bases.*

Algorithm 2 MergeBases($\lambda_1, B_1, \lambda_2, B_2$)

while $B_1 \neq B_2$ **do**
 Choose $i \in B_1 \setminus B_2$ and $j \in B_2 \setminus B_1$ such that
 both $B_1 - i + j$ and $B_2 - j + i$ are bases.
 With probability $\lambda_1/(\lambda_1 + \lambda_2)$ set $B_2 \leftarrow B_2 - j + i$;
 otherwise set $B_1 \leftarrow B_1 - i + j$.
end while
return B_1 .

Algorithm 3 Randomized swap rounding

Input: A decomposition as in (5).
 $C_1 \leftarrow B_1$ and $\gamma_1 \leftarrow \lambda_1$.
for $k = 1, \dots, m - 1$ **do**
 $C_{k+1} \leftarrow \text{MergeBases}(\gamma_k, C_k, \lambda_{k+1}, B_{k+1})$.
 $\gamma_{k+1} \leftarrow \gamma_k + \lambda_{k+1}$.
end for
return C_m .

The probability orientation in Algorithm 2 is important. Updating B_2 changes the weighted point by $\lambda_2(\mathbf{e}_i - \mathbf{e}_j)$, whereas updating B_1 changes it by $-\lambda_1(\mathbf{e}_i - \mathbf{e}_j)$. The displayed probabilities make the conditional expected displacement zero.

Theorem 4.2 (Swap rounding [7, 8]). *Algorithm 3 returns a random base R satisfying*

$$\mathbb{E}[\chi^R] = x, \quad \mathbb{E}[f(R)] \geq F(x).$$

Given the decomposition, it uses at most $(m - 1)r(N)$ symmetric exchanges. A direct search for exchange partners takes $O(mr(N)^2)$ independence-oracle calls.

Proof. One elementary merge step changes the current weighted point in one of the two ways

$$+\lambda_2(\mathbf{e}_i - \mathbf{e}_j) \quad \text{or} \quad -\lambda_1(\mathbf{e}_i - \mathbf{e}_j),$$

with probabilities $\lambda_1/(\lambda_1 + \lambda_2)$ and $\lambda_2/(\lambda_1 + \lambda_2)$, respectively. Its conditional expected displacement is zero. Thus the evolving weighted point satisfies Lemma 2.1. When all bases have merged, its value is χ^R , proving the marginal and objective guarantees.

Each exchange increases $|B_1 \cap B_2|$ by one, so a merge takes at most $r(N)$ exchanges. For a chosen i , testing the at most $r(N)$ candidates j finds a symmetric exchange partner, which gives the oracle bound. $\square$

	Randomized pipage rounding	Randomized swap rounding
Input form	Rank-inequality point $x \in B(M)$	Convex combination of bases
Primitive	Maximal feasible swap line	Symmetric basis exchange
Progress	Move to a proper subspace	Increase the overlap of two bases
Shared proof	Mean-preserving swaps and convexity of F	

5 Negative Correlation and Concentration

The same rounding distributions satisfy substantially more than correct marginals.

Definition 5.1 (Negative cylinder inequalities). For the output R , write $X_i = \mathbf{1}[i \in R]$ and $x_i = \mathbb{E}[X_i]$. The variables obey the *negative cylinder inequalities* if, for every $T \subseteq N$,

$$\mathbb{E}\left[\prod_{i \in T} X_i\right] \leq \prod_{i \in T} x_i, \quad \mathbb{E}\left[\prod_{i \in T} (1 - X_i)\right] \leq \prod_{i \in T} (1 - x_i). \quad (6)$$

Theorem 5.2 (Negative correlation [8]). *The output of randomized pipage rounding or randomized swap rounding satisfies (6).*

Proof. Fix $T \subseteq N$ and define

$$f_T^0(S) = \mathbf{1}[S \cap T \neq \emptyset], \quad f_T^1(S) = \mathbf{1}[T \not\subseteq S].$$

The first is a coverage function and is submodular. The second equals $f_T^0(N \setminus S)$ and is also submodular. Their multilinear extensions are

$$F_T^0(y) = 1 - \prod_{i \in T} (1 - y_i), \quad F_T^1(y) = 1 - \prod_{i \in T} y_i.$$

The rounding distribution is independent of the objective, so the no-loss guarantee applies to both functions. Hence

$$\begin{aligned} 1 - \mathbb{E}\left[\prod_{i \in T} (1 - X_i)\right] &= \mathbb{E}[f_T^0(R)] \geq F_T^0(x), \\ 1 - \mathbb{E}\left[\prod_{i \in T} X_i\right] &= \mathbb{E}[f_T^1(R)] \geq F_T^1(x). \end{aligned}$$

Rearranging gives (6). □

Negative cylinder inequalities are precisely what the moment-generating function proof of Chernoff bounds needs; full independence is unnecessary [9].

Corollary 5.3 (Chernoff bounds). *Let $a_i \in [0, 1]$, $Z = \sum_{i \in N} a_i X_i$, and $\mu = \mathbb{E}[Z] = \sum_i a_i x_i$. Then*

$$\begin{aligned} \Pr[Z \geq (1 + \delta)\mu] &\leq \left(\frac{e^\delta}{(1 + \delta)^{1 + \delta}}\right)^\mu \leq \exp\left(-\frac{\mu\delta^2}{2 + \delta}\right) & (\delta \geq 0), \\ \Pr[Z \leq (1 - \delta)\mu] &\leq \exp\left(-\frac{\mu\delta^2}{2}\right) & (0 \leq \delta \leq 1). \end{aligned}$$

Example 5.4 (Rounding a fractional spanning tree). For a connected graph $G = (V, E)$, the graphic-matroid base polytope is

$$\left\{x \in \mathbb{R}_{\geq 0}^E : x(E) = |V| - 1, \quad x(E(S)) \leq |S| - 1 \text{ for } \emptyset \neq S \subseteq V\right\}.$$

Pipage or swap rounding returns a spanning tree R with edge marginals x_e . For every edge set $A \subseteq E$, Corollary 5.3 concentrates $|R \cap A|$ around $x(A)$. Thus the rounding preserves both the expected load and independent-sampling-style tail bounds, even though the edges of a spanning tree cannot be independent.

Summary. Submodularity makes the multilinear extension convex along swap directions. Both randomized pipage rounding and randomized swap rounding move only in such directions and preserve the current point in conditional expectation. This yields exact marginals, no loss in expected multilinear value, negative correlation, and Chernoff concentration.

References

- [1] A. A. Ageev and M. I. Sviridenko, “Approximation algorithms for maximum coverage and max cut with given sizes of parts,” in *Integer Programming and Combinatorial Optimization*, LNCS 1610, Springer, 1999, pp. 17–30. doi:10.1007/3-540-48777-8_2.
- [2] A. A. Ageev and M. I. Sviridenko, “Pipage rounding: A new method of constructing algorithms with proven performance guarantee,” *Journal of Combinatorial Optimization* 8 (2004), 307–328. doi:10.1023/B:JOCO.0000038913.96607.c2.
- [3] R. A. Brualdi, “Comments on bases in dependence structures,” *Bulletin of the Australian Mathematical Society* 1 (1969), 161–167. doi:10.1017/S000497270004140X.
- [4] G. Călinescu, C. Chekuri, M. Pál, and J. Vondrák, “Maximizing a submodular set function subject to a matroid constraint,” in *Integer Programming and Combinatorial Optimization*, LNCS 4513, Springer, 2007, pp. 182–196. doi:10.1007/978-3-540-72792-7_15.
- [5] G. Călinescu, C. Chekuri, M. Pál, and J. Vondrák, “Maximizing a monotone submodular function subject to a matroid constraint,” *SIAM Journal on Computing* 40 (2011), 1740–1766. doi:10.1137/080733991.
- [6] W. H. Cunningham, “Testing membership in matroid polyhedra,” *Journal of Combinatorial Theory, Series B* 36 (1984), 161–188. doi:10.1016/0095-8956(84)90023-6.
- [7] C. Chekuri, J. Vondrák, and R. Zenklusen, “Dependent randomized rounding via exchange properties of combinatorial structures,” in *Proceedings of the 51st Annual IEEE Symposium on Foundations of Computer Science*, 2010, pp. 575–584. doi:10.1109/FOCS.2010.60.
- [8] C. Chekuri, J. Vondrák, and R. Zenklusen, “Dependent randomized rounding for matroid polytopes and applications,” arXiv:0909.4348, 2009, revised 2018. doi:10.48550/arXiv.0909.4348.
- [9] A. Panconesi and A. Srinivasan, “Randomized distributed edge coloring via an extension of the Chernoff–Hoeffding bounds,” *SIAM Journal on Computing* 26 (1997), 350–368. doi:10.1137/S0097539793250767.
- [10] J. Vondrák, “Optimal approximation for the submodular welfare problem in the value oracle model,” in *Proceedings of the 40th Annual ACM Symposium on Theory of Computing*, 2008, pp. 67–74. doi:10.1145/1374376.1374389.